

Curriculum Vitae - Jesse van Remmerden

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Research Profile

Fourth-year PhD researcher at TU/e building **AI that plans and schedules**: reinforcement learning, generative (diffusion) models and language models applied to vehicle routing, factory scheduling and infrastructure maintenance. I take bets the field considers unlikely and make them work: the first job-shop scheduler trained from logged data instead of a simulator (*Machine Learning*, 2025) and its successor, which learns from *random* data yet beats simulator-trained methods (*TMLR*, 2026); and the first route planner that follows instructions written in plain English, satisfying them 98–99% of the time where existing solvers manage ~32% (under review). In every project, I built the training data, the automatic checker that decides whether a solution is correct, and the benchmark myself, and released the code.

Selected Research

DROVER: a route planner that follows instructions in plain English

2025 - Present

First author; under review. Method: language-conditioned discrete diffusion

- Existing neural routing solvers plan a tour but cannot take requests such as “visit 7 before 4” or “keep these three stops on one truck”. I built the first benchmark for this setting: 26+ request types, $\sim 3 \times 10^9$ possible phrasings, reference solutions from established solvers, and an automatic checker that verifies whether a proposed tour actually obeys the request.
- Designed a generative model (discrete diffusion) that reads the request through a frozen language model (Gemma 4; Qwen 3.5 and Ministral 3 also tested) and steers its solution toward it with a new guidance technique.
- Obeys the request in 98–99% of cases on tours of 50 to 1,000 stops, versus ~32% for the best existing neural solvers; 87–90% on request types never seen in training (existing solvers: 0%); and, without any request, it remains competitive with the best neural solvers on plain tour length (best at 100 stops).

CDQAC & Offline-LD: learning factory schedules from data, not simulators

2023 - 2026

First author; Machine Learning (2025), TMLR (2026). Method: offline RL for job-shop scheduling

- Job-shop scheduling assigns factory operations to machines so everything finishes as early as possible. Deep-RL schedulers needed millions of trial-and-error runs in a simulator; Offline-LD was the first to learn purely from a fixed set of past schedules, matching or beating the simulator-trained baseline from only 100 examples.
- CDQAC went further: trained only on schedules produced by *random* decisions, it matches or beats the best simulator-trained methods on four standard benchmark suites, scales to problems with ten times as many jobs as seen in training (beating industrial solvers), and needs just 10–25 training examples.
- Explained why it works: what matters is how much of the decision space the data covers, not how good the example schedules are (rank correlation –0.95), a practical rule for building training data for decision-making models.

MO-DCMAC: 50-year maintenance planning for Amsterdam's historic canal walls

2023 - 2025

With the City of Amsterdam; Neural Computing & Applications (2025). Method: multi-objective RL

- Turned the risk-scoring method Amsterdam's asset managers use in practice (FMECA) into the training objective, so the learned plan optimizes what they actually care about: cost and probability of collapse. Best plan in 5 of 6 test settings, up to 46% better than the rule-based strategies in use; the first deep multi-objective RL method of its kind applied to a real-world problem.

Education

PhD Candidate, Artificial Intelligence for Planning and Scheduling

Jan. 2023 - Jan 2027

Eindhoven University of Technology (TU/e), AI for Decision Making, Information Systems IE&IS

- Five first-author papers (three journal, one workshop, one under review); three open-source codebases.

MSc Artificial Intelligence

Feb. 2020 - Jun. 2022

Utrecht University

- *Cum Laude* (8.2/10); [thesis](#) 8.6/10 on reinforcement learning for a graph-search problem; Honours Young Innovators programme (Utrecht Honours College).

BSc Computer Science

Sept. 2016 - Oct. 2019

University of Amsterdam

- [Thesis](#) 9/10: detecting dolphins in video with deep learning (Faster R-CNN).

Publications

- van Remmerden, J., Bukhsh, Z., & Zhang, Y. Generalizing Beyond Suboptimality: Offline Reinforcement Learning Learns Effective Scheduling through Random Solutions. [Transactions on Machine Learning Research](#) (2026). [Code](#)
- van Remmerden, J., Bukhsh, Z., & Zhang, Y. Offline reinforcement learning for learning to dispatch for job shop scheduling. [Machine Learning](#) 114, 191 (2025). [Code](#)
- van Remmerden, J., Kenter, M., Roijers, D.M., Andriotis, C., Zhang, Y., & Bukhsh, Z. Deep multi-objective reinforcement learning for utility-based infrastructural maintenance optimization. [Neural Computing and Applications](#) 37, 24719–24742 (2025).
- van Remmerden, J., Wu, Y., Zhang, Y., & Bukhsh, Z. DROVER: Diffusion-Based Routing with Verbal Preferences. *Under review*.
- van Remmerden, J., Kenter, M., Roijers, D.M. *et al.* A Deep Multi-Objective Reinforcement Learning Approach for Infrastructural Maintenance Planning with Non-Linear Utility Functions. *Multi-Objective Decision Making Workshop* (2023).

Professional Experience

Blockchain Engineer, Kryha

Aug. 2018 - Jan. 2020

- Built proofs-of-concept with smart contracts, tokenization, cryptography, and federated learning.
- First prize, Odyssey Hackathon 2019 (Nature 2.0 track): a federated-learning, blockchain, and evolutionary-computing proof-of-concept built in one weekend.

Technical Skills

Programming: Python (8+ yrs.; PyTorch, PyTorch Lightning, JAX), C/C++, Rust, SQL, LaTeX.

Machine learning: Offline and online reinforcement learning (CQL, QR-DQN, SAC, multi-objective RL), discrete diffusion and classifier-free guidance, graph transformers, conditioning on frozen LLM encoders, reward/verifier/benchmark design, synthetic data generation.

Optimization & infrastructure: OR-Tools (CP-SAT), Gurobi, LKH; Linux/HPC (SLURM, Apptainer, Kubernetes/Run:AI), multi-GPU training on A100/H100/B200, Docker, Git, Weights & Biases.

Languages: Dutch (native), English (fluent, C1).

Additional

Positions: Co-author of the 2022 municipal election programme, Volt Amsterdam (2021); head of the Health, Welfare and Sports Committee, Jonge Democraten Amsterdam (2016 - 2017).

Interests: Running, history, politics, mathematics, gaming, baking and cooking.

References: Dr. Yingqian Zhang (PhD promotor, yqzhang@tue.nl) and Dr. Zaharah A. Bukhsh (daily supervisor, z.bukhsh@tue.nl), Information Systems, Eindhoven University of Technology.